recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

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Abstract—Evaluating whether post-training enhances a language model's ability to recover from intermediate reasoning errors remains challenging due to confounding variables in trajectory evaluation. In this paper, we introduce recovery_eval, a data-centric evaluation framework that pairs verifier-defined error states with prospectively matched reference control states using frozen structural covariates and append-only exposure governance. We validate the framework across 400 genuine neural rollouts generated by two released checkpoints (Qwen2.5-Math-1.5B Base and Instruct) on 20 synthetic arithmetic problems inspired by GSM8K reasoning under a corrected state-matched protocol (Protocol 2.0; 256-token budget). Under state-matched prefixes, neither model exhibits spontaneous error recovery ($p_{\mathrm{BR}} = p_{\mathrm{IR}} = 0.0000$). Instead, models exhibit continuation inertia: 98.5% of recovery rollouts (197/200) faithfully propagate the injected error downstream. Under an independent-rollout assumption, the exact one-sided 95% Clopper-Pearson upper bound is 1.49% ($n = 200$; two-sided interval $[0.00\%, 1.83\%]$); accounting for intra-cluster correlation across the 20 problem templates yields an effective one-sided sensitivity bound of 13.91% ($C = 20$; two-sided interval $[0.00\%, 16.84\%]$). On unperturbed control prefixes, both models achieve near-perfect accuracy (Base: 97.00% heuristic, 100.00% structural; Instruct: 100.00%), yielding an observed sample recovery contrast of $D = 0.0000$ under structural verification (where sample variance collapses due to boundary saturation across the 20 items) and $D = -0.0300$ ($SE_{\mathrm{cluster}} = 0.0164, p = 0.0828$) under heuristic parsing. The historical negative interaction ($D \approx -0.21$) was not reproduced under Protocol 2.0, highlighting the acute sensitivity of recovery estimates to prompt construction, generation budgets, and scoring procedures, and establishing the necessity of state-matched evaluation infrastructure for reasoning diagnostics.

Index Terms—Language Models, Mathematical Reasoning, Error Recovery, State-Matched Evaluation, Data-Centric AI, Benchmark Governance, Continuation Inertia.

I. Introduction

Large language models (LLMs) trained on mathematical reasoning tasks exhibit substantial improvements in step-by-step problem-solving [1], [2]. A foundational question in machine learning evaluation is whether post-training procedures (such as instruction tuning, process supervision, or reinforcement learning) instill structural intelligence that enables models to detect and recover from early intermediate arithmetic or logical errors [3], [4].

However, evaluating error recovery behavior is acutely vulnerable to evaluation artifacts. Naive comparisons between error-containing trajectories and unperturbed baselines routinely conflate trajectory depth, remaining solution length, token complexity, and prompt phrasing. Furthermore, tight generation limits induce differential truncation censoring, masking true baseline capabilities. In preliminary experiments using an unbalanced 64-token protocol (Protocol 1.0), an apparent negative interaction was observed ($D \approx -0.21$), suggesting that instruction tuning degraded recovery capability. However, Protocol 1.0 and Protocol 2.0 differed simultaneously across prompt lengths, generation budgets, and scoring rules, leaving the relative causal contributions of individual design choices confounded.

To resolve these ambiguities, we present recovery_eval, a reproducible data-centric framework that enforces prospective caliper matching on intermediate reasoning states and maintains end-to-end cryptographic provenance. Across 400 genuine neural rollouts evaluated under a length-balanced, state-matched protocol with an expanded 256-token budget (Protocol 2.0), we establish that:

1) Non-Reproduction of the Negative Interaction: Under Protocol 2.0, the historical negative interaction was not reproduced ($D = -0.0300, p = 0.0828$ under heuristic parsing; $D = 0.0000$ under structural parsing), demonstrating the sensitivity of recovery estimates to protocol design.
2) Continuation Inertia: Small (1.5B) language models do not spontaneously self-correct from injected intermediate errors (0/200 recoveries). In 98.5% of cases, they treat corrupted premises as authoritative and faithfully propagate errors downstream.
3) Finite-Sample Uncertainty Bounds: We establish that observing zero recoveries corresponds to a one-sided 95% upper bound of 1.49% under independent rollout assumptions ($n = 200$), but cluster-level modeling reveals an effective sensitivity upper bound of 13.91% ($C = 20$).
4) Evaluator Sensitivity: Heuristic last-token regex extractors artificially penalize base models by 3% due to trailing conversational explanations, whereas

structural parsing demonstrates 100% baseline accuracy for both models.

II. Related Work

Self-correction and backtracking in LLMs have attracted intense recent scrutiny [5]–[7]. Table I compares recovery_eval with existing paradigms.

TABLE I
Comparison with Existing Reasoning Evaluation Frameworks

Framework	State Matching	Exposure Ledger	Primitive Provenance
End-to-End GSM8K [1]	None	No	Final Answer Only
PRM800K [3]	Unmatched	Partial	Step Scores
STaR [4]	Unmatched	No	Filtered Traces
recovery_eval (Ours)	State-Matched	Append-Only LEDGER	Full BPE Token + Raw JSONL

Unlike end-to-end benchmarks, recovery_eval prospectively matches recovery states with control states prior to generation, eliminating prefix length and depth confounds [8]–[10].

III. Problem Definition & Estimand

Let s denote a partial reasoning trajectory prefix for an arithmetic problem instance $p \in \mathcal{P}$. A deterministic verifier $\mathcal{V}(s)$ audits intermediate step correctness. A recovery state s_R contains a verifier-identified arithmetic error in Step 1. A reference control state s_C represents an unperturbed valid Step 1 prefix for the same problem at an equivalent problem-solving horizon.

For a target policy π (e.g., Instruct) and baseline policy π_0 (e.g., Base), let $V(s) \in \{0, 1\}$ denote the binary correctness indicator of completing the solution correctly from prefix s. We define the matched recovery contrast D as:

$$D = \mathbb{E}[V_\pi(s_R) - V_{\pi_0}(s_R)] - \mathbb{E}[V_\pi(s_C) - V_{\pi_0}(s_C)] \quad (1)$$

IV. The recovery_eval Protocol

A. State Construction and Injected Perturbation

Recovery states s_R are constructed by introducing controlled single-step arithmetic perturbations into Step 1 prefixes. Reference control states s_C maintain unperturbed Step 1 calculations.

B. Prospective Caliper Matching

To guarantee statistical balance, each recovery state s_R is matched to a control state s_C over structural covariates (trajectory depth, remaining length, and token length) ensuring exact categorical matching on reasoning operation (chained integer multiplication). In Protocol 2.0, Control and Recovery prefixes have identical character and token lengths ($\Delta_{\text{chars}} = 0, \Delta_{\text{tokens}} = 0$).

C. Exposure & Provenance Governance

All evaluation items are governed by an append-only event ledger with SHA-256 parent hash chaining to eliminate data leakage. Raw rollout records store input token IDs, generated BPE token IDs, wall-clock generation durations, and exact model weight manifest digests.

V. Experimental Configuration

The empirical campaign evaluates 20 synthetic multi-step arithmetic problems inspired by GSM8K word problems ($N = 20$). For each problem, 1 matched recovery state and 1 matched control state are evaluated across 2 model checkpoints using 5 stochastic generation seeds ($S = 5$), yielding exactly 400 rollouts ($20 \times 2 \times 2 \times 5 = 400$).

TABLE II
Evaluation Design and Provenance Statistics

Attribute	Specification / Value
Base Model	Qwen/Qwen2.5-Math-1.5B (Rev 4a83ca6e)
Instruct Model	Qwen/Qwen2.5-Math-1.5B-Instruct (Rev aafeb0fc)
Hardware Platform	NVIDIA GeForce RTX 4090 (24GB VRAM, CUDA 12.4)
Execution Precision	Native float16 PyTorch 2.4.0
Generation Config	$T = 0.7$, Top-$p = 0.9$, max_new_tokens = 256
Total Rollouts	400 (200 Base, 200 Instruct)
Total Generated Tokens	11,917 tokens (Mean: 29.79 tok/rollout)
Inference Duration	265.40 seconds (45.05 tokens/sec)
Truncation Rate	0.00% (0 / 400 rollouts truncated)
EOS Termination Rate	100.00% (400 / 400 rollouts hit EOS)
Raw Dataset SHA-256	b8792f8df375fccdbd800d440cc884ef 2db2f860ff3533af01c70161a5cc1d15

VI. Empirical Results

A. Cluster-Robust Factorial Outcomes

Table III summarizes completion accuracies across all four experimental cells under both heuristic and structural verification regimes.

TABLE III
Protocol 2.0 Factorial Outcomes ($N = 400$ Rollouts, $C = 20$ Clusters)

State Condition	Base (π_0)	Instruct (π)	Difference (Δ)
Recovery States (s_R)	0.0000	0.0000	+0.0000
Control States (Heuristic)	0.9700	1.0000	+0.0300
Control States (Structural)	1.0000	1.0000	+0.0000
Matched Contrast (Heuristic)	–	–	−0.0300
Cluster SE / p-value	–	–	$SE = 0.0164, p = 0.0828$
95% Confidence Interval	–	–	$[-0.0643, +0.0043]$
Matched Contrast (Structural)	–	–	+0.0000 (Sample Null)
Cluster SE / p-value	–	–	Degenerate ($s = 0.0000$)

B. Non-Reproduction of Negative Interaction

Under Protocol 2.0, the historical negative interaction was not reproduced. Under the heuristic parser, $D = -0.0300$ ($SE_{\text{cluster}} = 0.0164, t = -1.831, p = 0.0828$). Because the 95% confidence interval ($[-0.0643, +0.0043]$) crosses zero, the effect is statistically indistinguishable

from zero at $\alpha = 0.05$. Under structural arithmetic parsing, the observed sample contrast is $D = 0.0000$ across all 20 problem clusters. We emphasize that the resulting empirical zero standard error ($SE = 0.0000$) is a descriptive consequence of ceiling (100%) and floor (0%) saturation across these specific 20 items, rather than proof of universal population equivalence. Because Protocol 2.0 simultaneously updated prefix matching, token budgets, and scoring rules, the design demonstrates the collective sensitivity of recovery metrics to evaluation parameters rather than attributing the historical estimate to a single isolated factor.

C. Independent Verifier Falsification

The 3% deficit in Base Control (97% vs. 100%) was audited trace-by-trace. In all three disputed rollouts (problems fresh_004, fresh_006, fresh_013 under seed 405), the Base model calculated the correct revenue $(180, 176, 360)$ but appended a post-solution explanation concluding in "-In step 3...". The heuristic extractor's last-token fallback rule parsed the step index 3 instead of the revenue answer. Under structural equivalence, Base Control accuracy is 100/100 (1.0000).

D. Behavioral Trace Analysis: Continuation Inertia

Across all 200 recovery rollouts:

- Direct Error Propagation (98.5%, 197/200): Models adopted the corrupted Step 1 quantity E and computed $4 \times E$.
- Attention-Slip Divergence (1.5%, 3/200): In gsm8k_fresh_004, the model bound the unit price token into the hourly rate.
- Zero Backtracking (0.0%, 0/200): Zero rollouts contained error-detection words (wait, actually, mistake).

E. Uncertainty Bounds for Zero Recovery

Across the 200 recovery rollouts, no successful recovery was observed (0/200). Rather than interpreting zero sample events as an assertion of universal inability to self-correct, we report exact Clopper-Pearson confidence limits with explicitly disclosed assumptions. Under an independent Bernoulli assumption ($n = 200$), the one-sided 95% upper bound is 1.49% (two-sided 95% interval $[0.00\%, 1.83\%]$) [11]. However, because rollouts are grouped into 20 problem clusters evaluated over 5 stochastic seeds, repeated observations within clusters violate independence. As a sensitivity benchmark treating the 20 problem clusters as the sampling unit ($C = 20$), the one-sided 95% upper bound is 13.91% (two-sided 95% interval $[0.00\%, 16.84\%]$), as summarized in Table IV.

TABLE IV
Clopper-Pearson Exact Confidence Bounds for Zero Observed Recovery ($k = 0$)

Aggregation Tier	Effective Sample (N)	One-Sided 95% Upper Bound	Two-Sided 95% Interval
Rollout Level (Total)	200	1.49%	[0.00%, 1.83%]
Model Level (Base / Inst)	100	2.95%	[0.00%, 3.62%]
Cluster Level (Sensitivity)	20	13.91%	[**0.00%**, **16.84%**]

F. Human Validation and Annotation Disclosure

To validate automated verification against human judgment, we constructed a stratified blinded packet of 40 rollouts across models and conditions, audited against a six-dimensional annotation rubric. Author inspection demonstrated concordance with our semantic equivalence verifier, confirming 0% recovery, 100% error propagation, and validating that base-model trailing explanations represent mathematically correct reasoning. To maintain scientific integrity, formal multi-rater crowdsourced inter-rater reliability has not been executed and is designated as pending future work for camera-ready revision.

VII. Limitations

- Model Scale: Evaluated on two 1.5B parameter checkpoints; behavior of larger models (7B–72B) or reasoning-specialized models (e.g., DeepSeek-R1) remains an open empirical question.
- Task Scope: Evaluated on 20 synthetic arithmetic continuation tasks; does not cover multi-branch logic, open-domain theorem proving, or complex algorithmic problem-solving.
- Clustered Sampling: Repeated generation seeds within problem clusters violate independent sampling assumptions, requiring sensitivity bounds rather than unconditional population guarantees.
- Decoding Scaffolding: Evaluated under unprompted nucleus sampling ($T = 0.7, p = 0.9$) without explicit reflection prompts (*"check your work"*) or external verifier tree search.
- Causal Boundaries: The assigned prefix condition is causally identified, but checkpoint comparisons remain observational.
- Human Validation Status: Formal independent crowdsourced adjudication with inter-rater reliability metrics remains pending.

VIII. Reproducibility & Artifact Availability

All code, raw evidence, and verification scripts are archived. The complete neural dataset 10_COMPLETE_QWEN_PROTOCOL2_ROLLOUTS.jsonl (SHA-256: b8792f8df375fccdbd800d440cc884ef2db2f860ff3533af01c70161a5cc1d15) and evaluation package are preserved in the repository.

IX. Conclusion

We presented recovery_eval, a state-matched evaluation framework for language-model reasoning trajectories. The historical negative interaction was not reproduced under Protocol 2.0, demonstrating that recovery estimates are highly sensitive to prompt construction, generation budgets, and scoring procedures, while establishing that 1.5B language models exhibit continuation inertia rather than intrinsic self-correction under the evaluated conditions.

References

[1] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, "Training verifiers to solve math word problems," in arXiv preprint arXiv:2110.14168, 2021.

[2] A. Yang, B. Zhang, B. Zheng, D. Liu, J. Zhou et al., "Qwen2.5-Math technical report: Toward open math large language models with mathematical reasoning capabilities," arXiv preprint arXiv:2409.12122, 2024.

[3] H. Lightman, V. Kosaraju, Y. Shen, G. Hase, P. Clark et al., "Let's verify step by step," in International Conference on Learning Representations (ICLR), 2024.

[4] E. Zelikman, Y. Wu, J. Mu, and N. D. Goodman, "STaR: Bootstrapping reasoning with reasoning," in Advances in Neural Information Processing Systems (NeurIPS), 2022.

[5] C. Snell, K. Lee, K. Xu, and S. Levine, "Scaling LLM test-time compute optimally can be more effective than scaling model parameters," arXiv preprint arXiv:2408.03314, 2024.

[6] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," International Conference on Learning Representations (ICLR), 2023.

[7] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in International Conference on Learning Representations (ICLR), 2024.

[8] P. R. Rosenbaum and D. B. Rubin, "The central role of the propensity score in observational studies for causal effects," Biometrika, vol. 70, no. 1, pp. 41–55, 1983.

[9] D. E. Ho, K. Imai, G. King, and E. A. Stuart, "Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference," Political Analysis, vol. 15, no. 3, pp. 199–236, 2007.

[10] P. C. Austin, "An introduction to propensity score methods for reducing the effects of confounding in observational studies," Multivariate Behavioral Research, vol. 46, no. 3, pp. 399–424, 2011.

[11] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," Biometrika, vol. 26, no. 4, pp. 404–413, 1934.